How to play Garden Guard

EE2028 Assignment 2 | STM32L4S5 | Updated 24 October 2026

Garden Guard runs on STM32. Defend the garden against zombies while allowing friendly humans to pass. Your laptop shows the original 128 by 64 pixel game. The board reads the button and sensors, calculates the score, and plays the buzzer.

Use the physical blue USER/B2 button. The keyboard and mouse do not control the game. In this guide, "hold fire" means leave the button alone while a human passes; it does not mean hold the button down.

Connect and open the game

1. Connect the board's ST-Link USB connector, labelled CN7, to your laptop using a USB data cable. Keep the board still in its intended upright position for a second or two after power-on or reset so it can establish its orientation.
2. Keep the buzzer on D6/PB1 with its existing power and ground. The damaged OLED is optional; use a Grove I2C socket if you connect it. Unplug USB before changing wiring.
3. In the Assignment 2 folder, open `tools` and double-click `live_display.command`. Alternatively, open a terminal in Assignment 2 and run:

   ```
   sh tools/live_display.command
   ```

4. Wait for the LIVE badge and the board's screen. Opening the viewer does not reset the board or start a new round.

The viewer targets 30 frames per second. A connected-board check measured about 29.5 displayed frames per second; the status line shows the current measured rate. This is a live mirror of the board. `--preview` shows offline artwork.

Learn the button timing

A double press is two accepted presses **strictly less than 500 ms apart**. Exactly 500 ms is not a double. Use two distinct quick taps, roughly 100 to 300 ms apart, releasing the button between them. A 30 ms debounce filter rejects contact bounce; extremely rapid taps may be treated as one press.

A single press is confirmed only after its 500 ms double-press window has expired. Allow about 0.75 to 1 second between intentional single presses. Tapping twice too quickly requests a mode change instead of two charges.

Current state	Single press	Double press	Board LED
Standby	Ignored	Start a battle round	Continuously on
Normal battle	Add energy; automatically fire if charged	Return to standby	One blink per second
Attacked battle	Ignored	Rescue before the deadline	Two blinks per second
Terminated	Ignored	Ignored	Off

After an ordinary win or loss, the board remains in battle mode. Its normal charging and mode controls still work, although the completed round no longer awards target points.

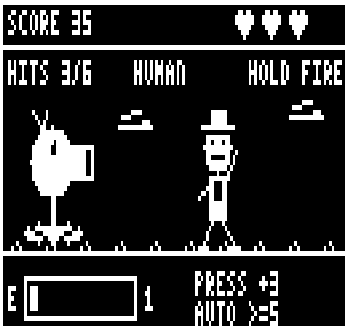
Start and read the screen

At standby, double press USER to begin. A fresh round starts with three hearts, zero points and zero zombie hits. Your goal is to hit six zombies before all three hearts are lost. Each target has a four-second period unless your shot ends that encounter sooner.

The screen shows your score, remaining hearts, energy and zombie-hit progress. Read the target label as well as its shape: `ZOMBIE` means prepare a shot; `HUMAN` and `HOLD FIRE` mean wait. When a target changes, check its identity again before pressing. A successful shot can bring the next target immediately.



Illustrative zombie scene



Illustrative human scene

These images are illustrative frames from the actual firmware renderer, not evidence of a live play-through.

Charge, shoot and score

There is no separate shoot button. Each confirmed single press adds three energy units, up to a capacity of ten. Whenever that addition gives at least five units, the board immediately fires one shot and spends five units.

Starting from empty, five separate singles produce this sequence:

Single press	Energy after charging	Automatic shot?	Energy left
1	3	No	3
2	6	Yes	1
3	4	No	4
4	7	Yes	2
5	5	Yes	0

Watch the energy bar rather than assuming every second press fires. For a first attempt, make well-separated singles while a zombie is displayed, then stop and read the new target after each shot. Leave room for the half-second input decision window instead of tapping at the very end of an encounter.

Outcome	Score change	Hearts
Hit a zombie	+10	Unchanged
Hit a zombie while IR aim is locked	+15 total	Unchanged
Allow a human to pass for its full four seconds	+5	Unchanged
Shoot a human	-5	Lose one
Let a zombie escape after four seconds	-5	Lose one

The reflective IR sensor supplies an optional aim-lock bonus. It detects a nearby object; it does not identify a human, zombie, red colour or green colour. The on-screen target decides what your shot hits. You can play without the IR bonus, and lack of aim lock never blocks automatic firing.

Warnings, attacks and rescue

A small `!` warning indicates an abnormal or unavailable sensor reading. It does not by itself mean game over. Sensor monitoring and UART reporting continue; normal telemetry resumes when all readings return to normal.

An upside-down attack is a separate state. The firmware compares gravity with the upright reference captured after reset and uses gyro rotation to support the detection. To try it, start in battle, turn the board smoothly over about 0.5 to 2 seconds, then hold it inverted. Avoid shaking it. A qualifying inverted position must persist for at least 350 ms.

The attacked screen replaces the target view with a rescue countdown. Double press USER promptly. Both taps must form a valid double, and its second press must occur before ten seconds have elapsed since the attack. Exactly 10,000 ms is too late. A single press has no effect here.

A rescue keeps your score, lives, energy and current target, but restarts that target's four-second timer. Return upright before attempting another attack. This rearms detection; staying inverted should not repeatedly trigger new attacks. Physical flip, rescue and timeout checks remain pending, so rehearse these actions on the actual board before relying on them in an assessment demo.

Win, lose and restart

Six zombie hits win the round. Losing all three hearts produces an ordinary round defeat. To play again after either result, double press to reach standby, wait briefly, then double press again to start battle. Score, hearts and hit count reset on battle entry. Energy is preserved across these mode changes.

A missed rescue deadline causes irreversible termination. The board finishes its final notification, stops UART communication and normal processing, and waits for a hardware reset. More USER presses cannot rescue it. Press the board's RESET button to restart in standby with zero energy, then keep it upright while the new orientation reference is acquired. The laptop retains the final image, shows `SESSION ENDED`, and stops debugger polling. After resetting the board, click `Reconnect after reset` to resume the laptop display.

If something looks wrong

- **No board or disconnected:** check the CN7 data cable and board power. Press `Reconnect` in the window, or close and reopen the launcher. A disconnected viewer marks the last image as paused.
- **Probe already in use:** close other ST-Link debuggers, programmers and old viewer windows, then reconnect. Keep one laptop viewer connected at a time.
- **Firmware mismatch:** the viewer checks the board against the local build. Use the matching project firmware and build instructions in the README.
- **Unexpected standby:** two intended charges may have become a double press. Space singles farther apart and double press once to re-enter battle.
- **No shot:** the first charge from empty leaves only three energy units. Check the bar and make the next single only when shooting is appropriate.
- **Blank or distorted OLED:** use the laptop screen. Damage to that panel does not prevent the laptop mirror or D6 buzzer from operating.

Escape or the window close button closes the viewer. The board continues running, so use standby when you want to stop playing without resetting it.